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Ex.No.4: OFDMA Resource Block Allocation

AIM: To simulate and analyze OFDMA (Orthogonal Frequency Division Multiple Access)

resource block allocation and evaluate system performance in terms of resource utilization and

user throughput using MATLAB.

Objective

1. To determine the Number of Resource Blocks (RBs) allocated in an OFDMA system

based on available bandwidth and system configuration.

2.

To evaluate the Resource Block Utilization (%) by analyzing how efficiently available

RBs are assigned to active users.

3.

To calculate the User Throughput (Mbps) based on allocated resource blocks and

modulation efficiency in the OFDMA system.

Procedure

1. Define the system bandwidth and calculate the number of available resource blocks

(RBs) based on RB size (e.g., 180 kHz per RB).

2. Identify the number of active users in the OFDMA system for resource sharing.

3. Allocate resource blocks to each user based on a predefined or equal distribution

strategy.

4. Compute Resource Block Utilization (%) using the ratio of allocated RBs to total RBs.

5. Assign modulation efficiency values (e.g., QPSK, 16-QAM, 64-QAM) for each user.

6. Calculate User Throughput (Mbps) using allocated RBs and modulation efficiency.

7. Plot graphs for RB allocation and utilization/throughput analysis using MATLAB.

8. Analyze results to evaluate spectrum efficiency and system performance.

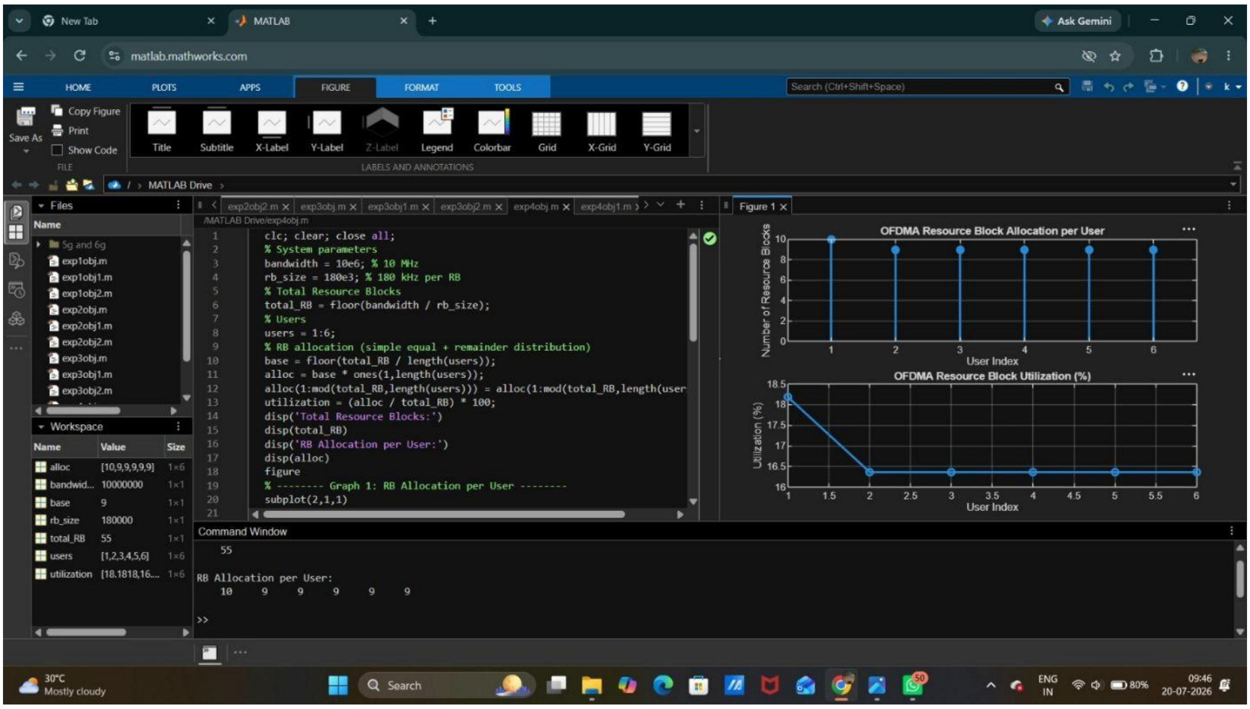
Objective1:

Program:

```
clc; clear; close all;

% System parameters

bandwidth = 10e6; % 10 MHz

rb_size = 180e3; % 180 kHz per RB

% Total Resource Blocks

total_RB = floor(bandwidth / rb_size);

% Users

users = 1:6;

% RB allocation (simple equal + remainder distribution)

base = floor(total_RB / length(users));

…xlabel('User Index')

ylabel('Number of Resource Blocks')

grid on

% -------- Graph 2: RB Utilization --------

subplot(2,1,2)

plot(users, utilization, '-o', 'LineWidth', 2)

title('OFDMA Resource Block Utilization (%)')

xlabel('User Index')

ylabel('Utilization (%)')
```

grid on

Output:



Objective2:

Program:

clc; clear; close all;

% Total system resource blocks

total_RB = 50;

% Number of active users

users = 1:5;

% RB allocation to users (example scenario)

alloc = [12 10 9 8 11];

% Resource Block Utilization (%)

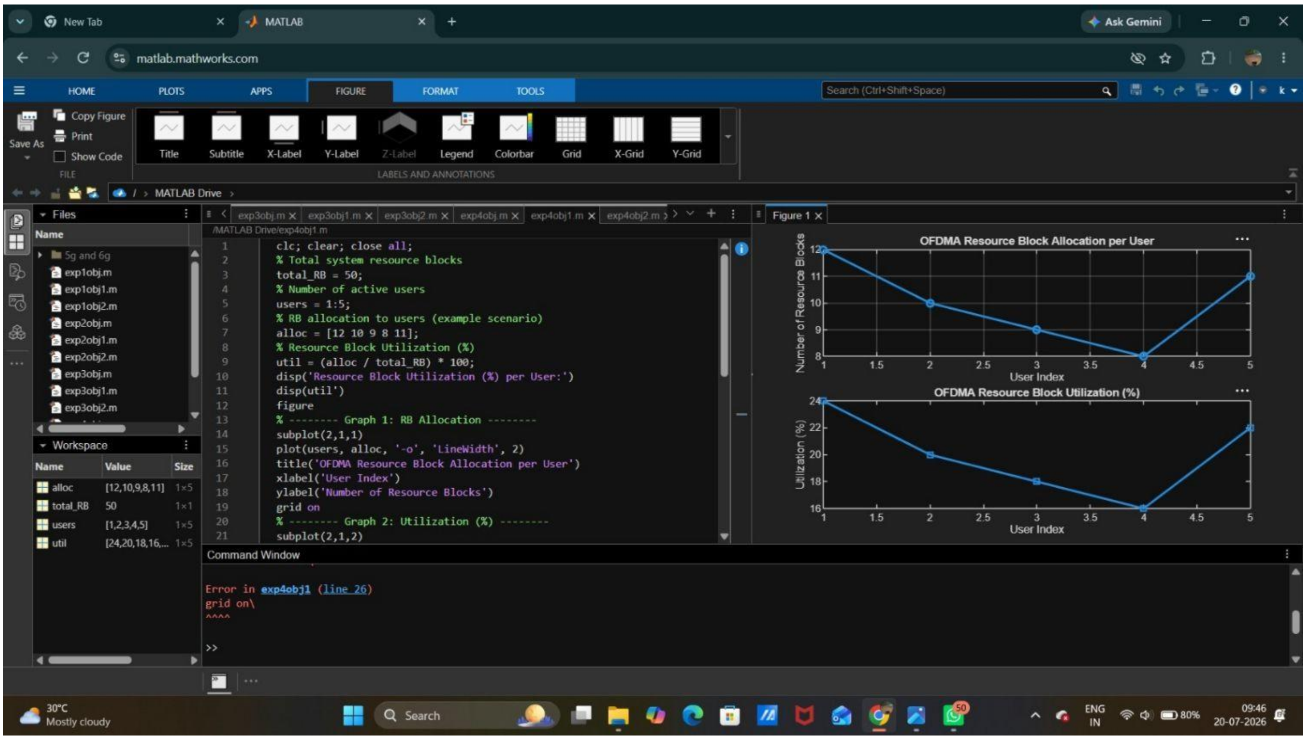
util = (alloc / total_RB) * 100;

disp('Resource Block Utilization (%) per User:')

…xlabel('User Index')

```
ylabel('Number of Resource Blocks')
grid on
% -------- Graph 2: Utilization (%) --------
subplot(2,1,2)
plot(users, util, '-s', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on
```

Output:



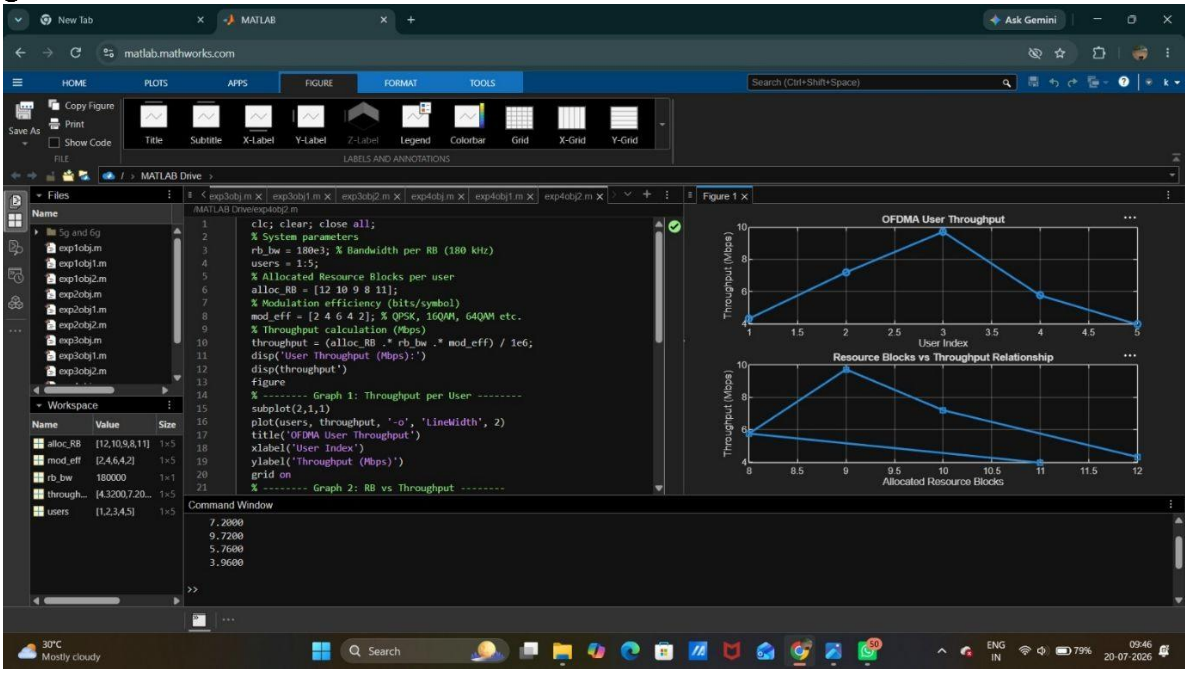
Objective3:

Program:

```
clc; clear; close all;
% System parameters
```

```
rb_bw = 180e3; % Bandwidth per RB (180 kHz)
users = 1:5;
% Allocated Resource Blocks per user
alloc_RB = [12 10 9 8 11];
% Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
% Throughput calculation (Mbps)
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
…xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
% -------- Graph 2: RB vs Throughput --------
subplot(2,1,2)
plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
title('Resource Blocks vs Throughput Relationship')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
```

grid on



Result:

1. The Resource Block (RB) allocation was successfully simulated for multiple users in the

OFDMA system, demonstrating how available spectrum is distributed among active

users.

2. The Resource Block Utilization (%) analysis showed efficient usage of system resources,

indicating how effectively the available RBs are assigned to users.

3. The User Throughput (Mbps) was successfully calculated based on RB allocation and

modulation efficiency, showing that higher allocated RBs and higher modulation

schemes result in increased throughput.